

AIMS
Airbus Material Specification

**Honeycomb Cores, N-636 Polyamide Paper (meta and para-aramid),
Phenolic Resin Impregnated
Over-expanded**

Material Specification

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1 Scope

This Material Specification defines the requirements of N-636 polyamide paper (meta and para-aramid) impregnated with phenolic resin in over-expanded configuration in a cured state for aerospace application. Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

Core material for the different types production of light-weight sandwich components, temperature range – 55°C to 135 °C.

Materials to be delivered according to ABS5670.

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ABS5670	Honeycomb cores, N-636 Polyamide paper (meta and para-aramid), Phenolic resin impregnated, over-expanded. Semi finished Product Standard.
AIMS 11-01-000 Part 1	Honeycomb Cores, Phenolic Resin Impregnated. General Requirements. Technical Specification (TS)
AIMS 11-01-000 Part 2	Honeycomb Cores, Phenolic Resin Impregnated. Polyamide paper / Hexagonal / Over-expanded / Flexible. Qualification Programme - Release Testing
EN2743	Aerospace series, Standard procedure for conditioning prior to testing
ASTM-C273	Shear test in flatwise plane of flat sandwich constructions or sandwich cores
ASTM-D 374	Standard test methode for thickness of solid electrical insulation
ASTM-D 646	Standard Test Method for Grammage of Paper and Paperboard (Mass Per Unit Area)

4 Technical requirements

4.1 General

Specification values are minimum individual except where stated

4.1.1 Material description

see table 1

Table 1: Material description

Characteristic	Requirements
Material type	Honeycomb core, N636 polyamide paper (meta and para-aramid), phenolic resin impregnated
Configuration	Over-expanded
State of material	cured
Material code	AIMS11-01-007B
Density	See table 2a

4.1.2 Worklife and storage conditions

Worklife at max. 30 °C/75 % RH. shall be at least 12 months.

4.1.3 Processing conditions

The performance of all grades of honeycomb covered by this MS shall not be reduced by:

- expose to ≤ 248 °C for short periods (e.g. heat forming)
- autoclave or press curing processes with adhesives and/or prepregs at ≤ 185 °C.

4.1.4 Hazardous Substances

The relevant local health/safety and environmental regulations shall be fulfilled.

4.2 Specific

4.2.1 Physical-chemical properties

See tables 2a and 2b

4.2.2 Mechanical properties

See table 3

4.2.3 Burning behaviour

See table 4

Table 2a: Physical properties

No.	Properties	Test method	Units	Requirements (nominal values)
Material code / grade				AIMS11-01-007B
1	Cell size ⁴⁾	TS Part 1 chapter: 5.3.1.5.2	mm	4.8
2	Theoretical paper thickness	TS Part 1 chapter: 5.3.1.5.3	µm	36
3	Density	TS Part 1 chapter: 5.3.1.5.5	kg/m ³	32
4a	Moisture absorption ³⁾	TS Part 1 chapters: 5.3.1.5.6 5.3.1.5.6	%	<5 ¹⁾
4b				<6 ²⁾
1) Moisture absorption at 70 °C/85 % R.H. 2) Moisture absorption 24 hours in water at R.T. (alternative method, see AIMS 11-01-000 Part 1). 3) Maximum individual values. 4) Cell size before over-expansion.				

Table 2b: Physical properties (cont.): Paper properties for information.

Properties		Test method	Units	Values based on qualification data
Material code / grade				AIMS11-01-007B
1	Theoretical paper thickness	TS Part 1 chapter: 5.3.1.5.3	μm	36
2	Real paper thickness ¹⁾			42.6
3	Areal weight	ASTM-D646	osy	0.9 ± 0.07
			g/m²	30.5 ± 2.4
1) Average values obtained in the qualification program process				

Table 3: Mechanical properties

No.	Properties	Test method	Test temp. [°C]	Units	Requirements	
Material code / grade					AIMS11-01-007B	
1a	Compression strength	ASTM-C365 + TS Part 1 chapter: 5.3.1.6.1	23 ¹⁾	MPa	0.78	
1b			70 ¹⁾		0.70	
1c			135 ¹⁾		0.62	
1d			180 ¹⁾		0.56	
1e			70 ²⁾		0.68	
2a	L-direction	ASTM-C273 + TS Part 1 chapter: 5.3.1.6.2	23 ¹⁾		0.38	
2b			70 ¹⁾		0.34	
2c			135 ¹⁾		0.32	
2d			70 ²⁾		0.34	
2e	Shear strength		23 ¹⁾		0.52	
2f	W-direction		70 ¹⁾		0.49	
2g			135 ¹⁾		0.45	
2h			70 ²⁾		0.49	
2j			L-direction		23 ¹⁾	22
2k	70 ¹⁾		20			
2l	135 ¹⁾		18			
2m	Shear modulus		70 ²⁾		20	
2n	W-direction		23 ¹⁾		46	
2o			70 ¹⁾		43	
2p			135 ¹⁾		40	
2q			70 ²⁾		43	
1) Dry, predrying for 1 h at 80 °C (according to EN 2743). 2) Wet 70 °C/85 % R.H. according to AIMS 11-01-000 Part 2.						

Table 4: Burning behaviour

No.	Properties			Test method	Unit	Max. Average Values
Material code / grade						AIMS11-01-007B
1a	Flammability vertical 60s flaming	Burn length		TS Part 1 chapter: 5.3.1.7	mm	75
1b		After flame time			s	5
1c		After flame time of drips			s	0
2	Max. specific optical smoke density	F ¹⁾		TS Part 1 chapter: 5.3.1.7	-	50
3a	Max. concentration of smoke gas components	F ¹⁾	HCN	TS Part 1 chapter: 5.3.1.7	ppm	20
3b			CO			500
3c			NO/NO ₂			20
3d			SO ₂ /H ₂ S			20
3e			HF			10
3f			HCl			10
4a	Heat release rate max. within 5 min. ²⁾			TS Part 1 chapter: 5.3.1.7	kw/m ²	-
4b	Heat release at 2 min ²⁾				kw _{xmin} /m ²	-
1) F = Flaming Mode						
2) Heat release testing shall be done within the qualification for low density honeycomb for interior parts only ; specimen and requirements shall be defined during qualification						

5 Designation

Refer to the relevant ABS

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS 11-01-000.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 07/ 2003		New Standard